



## (51) International Patent Classification:

C07D 237/16 (2006.01) A61K 31/53 (2006.01)

C07D 405/04 (2006.01) A61K 31/501 (2006.01)

## (21) International Application Number:

PCT/IN2025/050013

## (22) International Filing Date:

06 January 2025 (06.01.2025)

## (25) Filing Language:

English

## (26) Publication Language:

English

## (30) Priority Data:

202441001088 05 January 2024 (05.01.2024) IN

## (71) Applicant: MSN LABORATORIES PRIVATE LIMIT-

ED, R&amp;D CENTER [IN/IN]; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN).

## (72) Inventor; and

(71) Applicant: SRINIVASAN, Thirumalai Rajan [IN/IN]; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN).

(72) Inventors: SAGYAM, Rajeshwar Reddy; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN). KAMMARI, Balraju; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN). GARAI, Abhijit; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN). EDULAKANTI, Jhansi; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN). DODLE, Beerappa; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275 to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN).

## (74) Common Representative: SRINIVASAN, Thirumalai

Rajan; Plot No. 12, Phase-IV, Sy No. 119 to 140, 258, 275

to 280, IDA, Pashamylaram (Vill), Patancheru (Mdl), Sangareddy (Dist), Telangana (State), Hyderabad 502307 (IN).

## (81) Designated States (unless otherwise indicated, for every kind of national protection available):

AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MU, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

## (84) Designated States (unless otherwise indicated, for every kind of regional protection available):

ARIPO (BW, CV, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SC, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

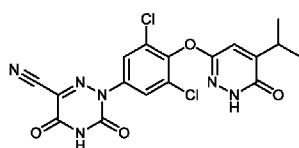
## Declarations under Rule 4.17:

- as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))
- of inventorship (Rule 4.17(iv))

## Published:

- with international search report (Art. 21(3))
- before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))
- in black and white; the international application as filed contained color or greyscale and is available for download from PATENTSCOPE

## (54) Title: SOLID-STATE FORMS OF RESMETIROM AND PROCESSES FOR PREPARATION THEREOF



Formula-1.

(57) Abstract: The present application relates to a various novel crystalline forms of 2-[3,5-dichloro-4-(5-isopropyl-6-oxo-1,6-dihydropyridazin-3-yloxy)phenyl]-3,5-dioxo-2,3,4,5-tetrahydro [1,2,4] triazine-6-carbonitrile represented by the following structural formula-1, which is referred to as Resmetirom. The present application also relates to process for the preparation of various novel crystalline polymorphic forms of compound of formula-1.



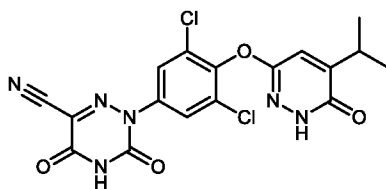
## Solid-state forms of Resmetirom and processes for preparation thereof

### Related Applications:

This application claims the benefit of priority to our Indian patent application number 202441001088 filed on Jan 05, 2024, disclosures of all of which is incorporated by reference in their entirety.

### Field of the invention:

The present application relates to a novel crystalline forms of 2-[3,5-dichloro-4-(5-isopropyl-6-oxo-1,6-dihydropyridazin-3-yloxy)phenyl]-3,5-dioxo-2,3,4,5-tetrahydro[1,2,4]triazine-6-carbonitrile represented by the following structural formula-1, which is referred to as Resmetirom.



Formula-1.

### Background of the invention:

2-[3,5-Dichloro-4-(5-isopropyl-6-oxo-1,6-dihydropyridazin-3-yloxy)phenyl]-3,5-dioxo-2,3,4,5-tetrahydro[1,2,4]triazine-6-carbonitrile or 2-[3,5-Dichloro-4-((6-oxo-5-(propan-2-yl)-1,6-dihydropyridazin-3-yl)oxy)phenyl]-3,5-dioxo-2,3,4,5-tetrahydro-1,2,4-triazine-6-carbonitrile, commonly known as Resmetirom, is approved by the US FDA under the brand name REZDIFFRA. It is a thyroid hormone receptor-beta (THR-beta) agonist indicated, in conjunction with diet and exercise, for the treatment of adults with noncirrhotic nonalcoholic steatohepatitis (NASH) with moderate to advanced liver fibrosis (stages F2 to F3 fibrosis).

(1/4)

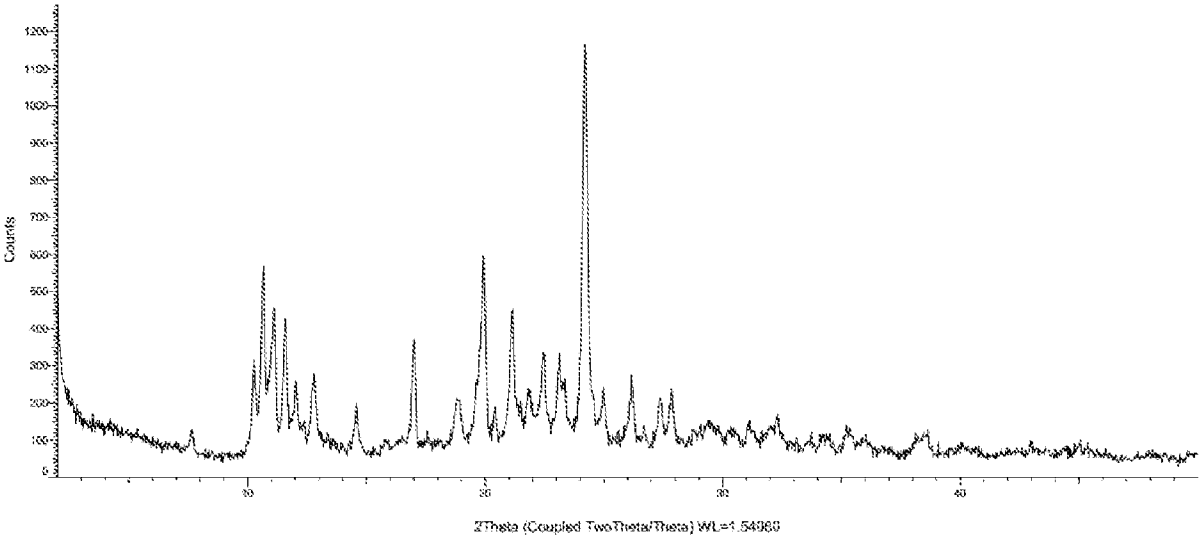


Figure-1

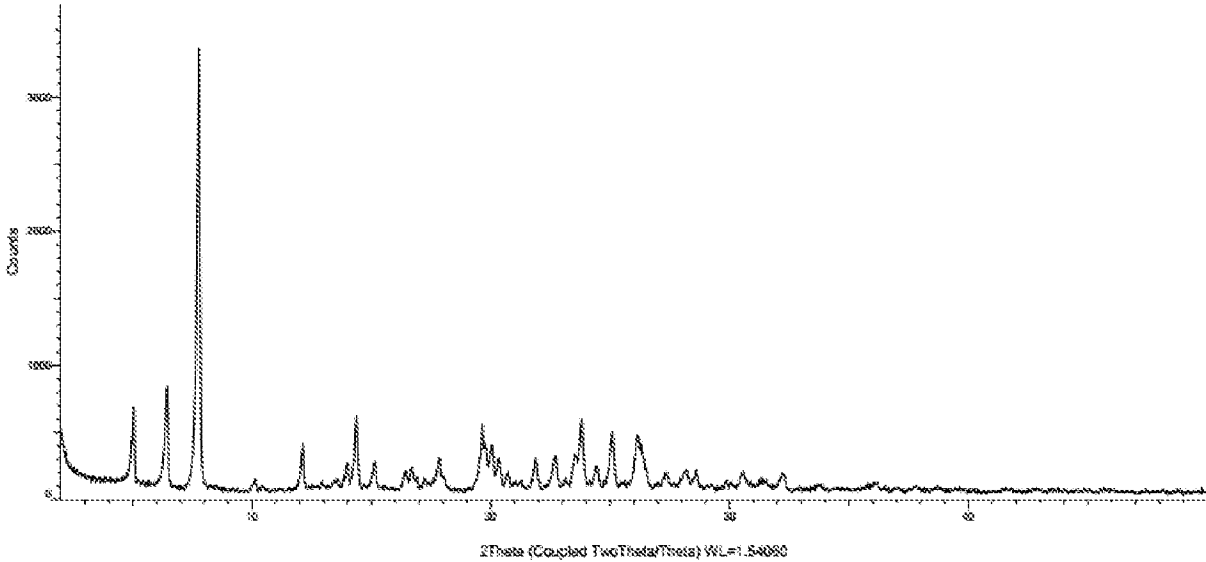


Figure-2